Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

Class:	**TE-AI&DS**	**Semester:**	**V**
Course Code:	**2015111**	**Course Name:**	**Statistics for Machine Learning and Data Science Lab**

Name of Student:	Kunal Yadav
Roll No. :	64
Experiment No.:	**9**
Title of the Experiment:	**Evaluate and Interpret Machine Learning Models using Validation and Explainability Technique**
Date of Performance:	**29/09/26**
Date of Submission:	**06/10/26**

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty :

Signature :

Date :

Aim: To apply model evaluation, Bayesian inference concepts, and explainable AI techniques to interpret and understand machine learning predictions on the Pima Indians Diabetes Dataset.

Objectives

After completing this experiment, the student will be able to:

1. Apply model evaluation and Bayesian concepts to analyze the reliability of machine learning predictions.
2. Apply explainable AI techniques to interpret model predictions and identify important features influencing the results.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- SHAP
- LIME

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for **768 female patients**.

The target variable is: Outcome = 0: Non-diabetic and Outcome = 1: Diabetic

Procedure

1. Load and preprocess the Pima Indians Diabetes Dataset.
2. Divide the dataset into training and testing sets.
3. Train a suitable classification model.
4. Evaluate the model using k-fold cross-validation.
5. Calculate suitable performance measures such as accuracy and F1-score.
6. Calculate the Brier Score and generate a calibration curve.
7. Explain selected model predictions using SHAP or LIME.
8. Identify the important features influencing the predictions.
9. Interpret and document the model evaluation and explainability results.

Description of the Experiment

A machine learning model should not be evaluated only on its performance on a single test split. Cross-validation provides a more reliable estimate of model performance.

The experiment follows: Dataset, Train Model, Cross-Validation, Performance Evaluation, Calibration Analysis, Explainable AI and Interpretation

The experiment combines three aspects:

- **Model Evaluation:** How well does the model perform?
- **Calibration:** How reliable are the predicted probabilities?
- **Explainability:** Why did the model make a particular prediction?

Detailed Description of Statistical Methods

k-Fold Cross-Validation

The dataset is divided into $k$subsets.

The model is trained on $k - 1$subsets and tested on the remaining subset.

This process is repeated $k$times.

The final performance is calculated using the average performance across all folds.

Bias-Variance Trade-Off

A model should balance:

- **Bias:** Error due to an overly simple model.
- **Variance:** Error due to a model being too sensitive to the training data.

A good model attempts to achieve a suitable balance between bias and variance.

Calibration Curve

A calibration curve compares:

$$Predicted\ Probability$$

with:

$$Observed\ Frequency$$

A well-calibrated model produces probabilities that closely match the actual frequency of the outcome.

Brier Score

The Brier Score measures the accuracy of probabilistic predictions:

$$BS = \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2$$

where:

- p_i= predicted probability

- y_i = actual outcome

A lower Brier Score generally indicates better probabilistic predictions.

SHAP

SHAP values explain the contribution of individual features to a model prediction.

They help answer:

Which features contributed to increasing or decreasing the predicted probability?

LIME

LIME explains an individual prediction by creating small variations around the observation and approximating the model locally using an interpretable model.

Code:-

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt


from sklearn.model_selection import train_test_split, StratifiedKFold, cross_validate
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression


from sklearn.metrics import (
    accuracy_score,
    f1_score,
    brier_score_loss,
```

```
    classification_report,
    confusion_matrix,
    ConfusionMatrixDisplay
)

from sklearn.calibration import calibration_curve

import shap
from lime.lime_tabular import LimeTabularExplainer

import warnings
warnings.filterwarnings("ignore")

url = "https://raw.githubusercontent.com/jbrownlee/Datasets/master/pima-indians-diabetes.data.csv"
columns = [
    "Pregnancies",
    "Glucose",
    "BloodPressure",
    "SkinThickness",
    "Insulin",
    "BMI",
    "DiabetesPedigreeFunction",
    "Age",
    "Outcome"
]
```

```
df = pd.read_csv(url, names=columns)


print("=" * 70)
print("PIMA INDIANS DIABETES DATASET")
print("=" * 70)


print("\nDataset Shape:")
print(df.shape)


print("\nFirst 5 Rows:")
print(df.head())


print("\nDataset Information:")
df.info()


print("\nClass Distribution:")
print(df["Outcome"].value_counts())


print("\nStatistical Summary:")
print(df.describe())



zero_as_missing = [
    "Glucose",
    "BloodPressure",
    "SkinThickness",
    "Insulin",
```

```
    "BMI"
]

df[zero_as_missing] = df[zero_as_missing].replace(0, np.nan)

print("\nMissing Values After Preprocessing:")
print(df.isnull().sum())

X = df.drop("Outcome", axis=1)
y = df["Outcome"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

print("\nTraining Set Size:", X_train.shape)
print("Testing Set Size:", X_test.shape)


pipeline = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler()),
    ("classifier", LogisticRegression(max_iter=1000))
```

```
])

cv = StratifiedKFold(
    n_splits=5,
    shuffle=True,
    random_state=42
)

scoring = {
    "accuracy": "accuracy",
    "f1": "f1"
}

cv_results = cross_validate(
    pipeline,
    X_train,
    y_train,
    cv=cv,
    scoring=scoring
)

cv_accuracy = cv_results["test_accuracy"]
cv_f1 = cv_results["test_f1"]

print("\n" + "=" * 70)
print("5-FOLD CROSS-VALIDATION")
print("=" * 70)
```

```
print("\nAccuracy:")
for i, score in enumerate(cv_accuracy, 1):
    print("Fold", i, ":", round(score, 4))

print("\nF1 Score:")
for i, score in enumerate(cv_f1, 1):
    print("Fold", i, ":", round(score, 4))

print("\nMean Accuracy:", round(cv_accuracy.mean(), 4))
print("Accuracy Std:", round(cv_accuracy.std(), 4))

print("\nMean F1 Score:", round(cv_f1.mean(), 4))
print("F1 Std:", round(cv_f1.std(), 4))
pipeline.fit(X_train, y_train)



y_pred = pipeline.predict(X_test)
y_prob = pipeline.predict_proba(X_test)[:, 1]
accuracy = accuracy_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)

print("\n" + "=" * 70)
print("TEST SET PERFORMANCE")
print("=" * 70)

print("\nAccuracy:", round(accuracy, 4))
```

```
print("F1 Score:", round(f1, 4))

print("\nClassification Report:")
print(classification_report(
    y_test,
    y_pred,
    target_names=["Non-diabetic", "Diabetic"]
))
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=["Non-diabetic", "Diabetic"]
)

disp.plot(cmap="Blues")
plt.title("Confusion Matrix")
plt.show()
brier = brier_score_loss(y_test, y_prob)

print("\n" + "=" * 70)
print("BRIER SCORE")
print("=" * 70)
```

```
print("\nBrier Score:", round(brier, 4))
prob_true, prob_pred = calibration_curve(
    y_test,
    y_prob,
    n_bins=10,
    strategy="uniform"
)

plt.figure(figsize=(7, 6))

plt.plot(
    prob_pred,
    prob_true,
    marker="o",
    linewidth=2,
    label="Logistic Regression"
)

plt.plot(
    [0, 1],
    [0, 1],
    "--",
    color="black",
    label="Perfect Calibration"
)

plt.xlabel("Mean Predicted Probability")
```

```
plt.ylabel("Observed Frequency")
plt.title("Calibration Curve")
plt.legend()
plt.grid(alpha=0.3)
plt.show()
print("\n" + "=" * 70)
print("SHAP EXPLAINABILITY")
print("=" * 70)

imputer = pipeline.named_steps["imputer"]
scaler = pipeline.named_steps["scaler"]
model = pipeline.named_steps["classifier"]

X_train_imputed = imputer.transform(X_train)
X_test_imputed = imputer.transform(X_test)

X_train_scaled = scaler.transform(X_train_imputed)
X_test_scaled = scaler.transform(X_test_imputed)

X_test_scaled_df = pd.DataFrame(
    X_test_scaled,
    columns=X.columns
)

explainer = shap.LinearExplainer(
    model,
    X_train_scaled
```

```
)

shap_values = explainer(X_test_scaled)
shap.summary_plot(
    shap_values,
    X_test_scaled_df,
    show=False
)

plt.title("SHAP Summary Plot")
plt.tight_layout()
plt.show()
mean_abs_shap = np.abs(shap_values.values).mean(axis=0)

feature_importance = pd.DataFrame({
    "Feature": X.columns,
    "Mean Absolute SHAP": mean_abs_shap
})

feature_importance = feature_importance.sort_values(
    "Mean Absolute SHAP",
    ascending=False
)

print("\nSHAP Feature Importance:")
print(feature_importance.to_string(index=False))
```

```
plt.figure(figsize=(8, 5))

plt.barh(
    feature_importance["Feature"],
    feature_importance["Mean Absolute SHAP"],
    color="steelblue"
)

plt.xlabel("Mean Absolute SHAP Value")
plt.ylabel("Feature")
plt.title("SHAP Feature Importance")

plt.gca().invert_yaxis()

plt.tight_layout()
plt.show()
sample_index = 0

print("\n" + "=" * 70)
print("INDIVIDUAL PATIENT PREDICTION")
print("=" * 70)

print("\nPatient Features:")
print(X_test.iloc[[sample_index]])

print("\nActual Outcome:",
      y_test.iloc[sample_index])
```

```
print("Predicted Outcome:",
      y_pred[sample_index])

print("Predicted Probability:",
      round(y_prob[sample_index], 4))
print("\nGenerating SHAP individual explanation...")

shap.plots.waterfall(
    shap_values[sample_index],
    max_display=10
)

plt.show()
print("\n" + "=" * 70)
print("LIME EXPLAINABILITY")
print("=" * 70)

lime_explainer = LimeTabularExplainer(
    X_train_scaled,
    feature_names=X.columns.tolist(),
    class_names=["Non-diabetic", "Diabetic"],
    mode="classification",
    random_state=42
)

lime_exp = lime_explainer.explain_instance(
```

```
    X_test_scaled[sample_index],
    model.predict_proba,
    num_features=len(X.columns)
)

print("\nLIME Explanation:")

for feature, weight in lime_exp.as_list():
    print(
        feature,
        ":",
        round(weight, 4)
    )
results = pd.DataFrame({
    "Metric": [
        "5-Fold Mean Accuracy",
        "5-Fold Accuracy Std",
        "5-Fold Mean F1",
        "5-Fold F1 Std",
        "Test Accuracy",
        "Test F1",
        "Brier Score"
    ],

    "Value": [
        cv_accuracy.mean(),
        cv_accuracy.std(),
```

```
        cv_f1.mean(),
        cv_f1.std(),
        accuracy,
        f1,
        brier
    ]
})

print("\n" + "=" * 70)
print("FINAL RESULTS")
print("=" * 70)

print(results.to_string(index=False))

print("\n" + "=" * 70)
print("TOP 5 IMPORTANT FEATURES")
print("=" * 70)

for rank, (_, row) in enumerate(
    feature_importance.head(5).iterrows(),
    1
):
    print(
        rank,
        ".",
        row["Feature"],
        "-> SHAP:",
```

```
        round(row["Mean Absolute SHAP"], 4)
    )
print("\n" + "=" * 70)
print("CONCLUSION")
print("=" * 70)

print("""
The Pima Indians Diabetes Dataset was used to train a
Logistic Regression classification model.

The model was evaluated using 5-fold stratified
cross-validation, accuracy, F1-score, Brier Score,
and a calibration curve.

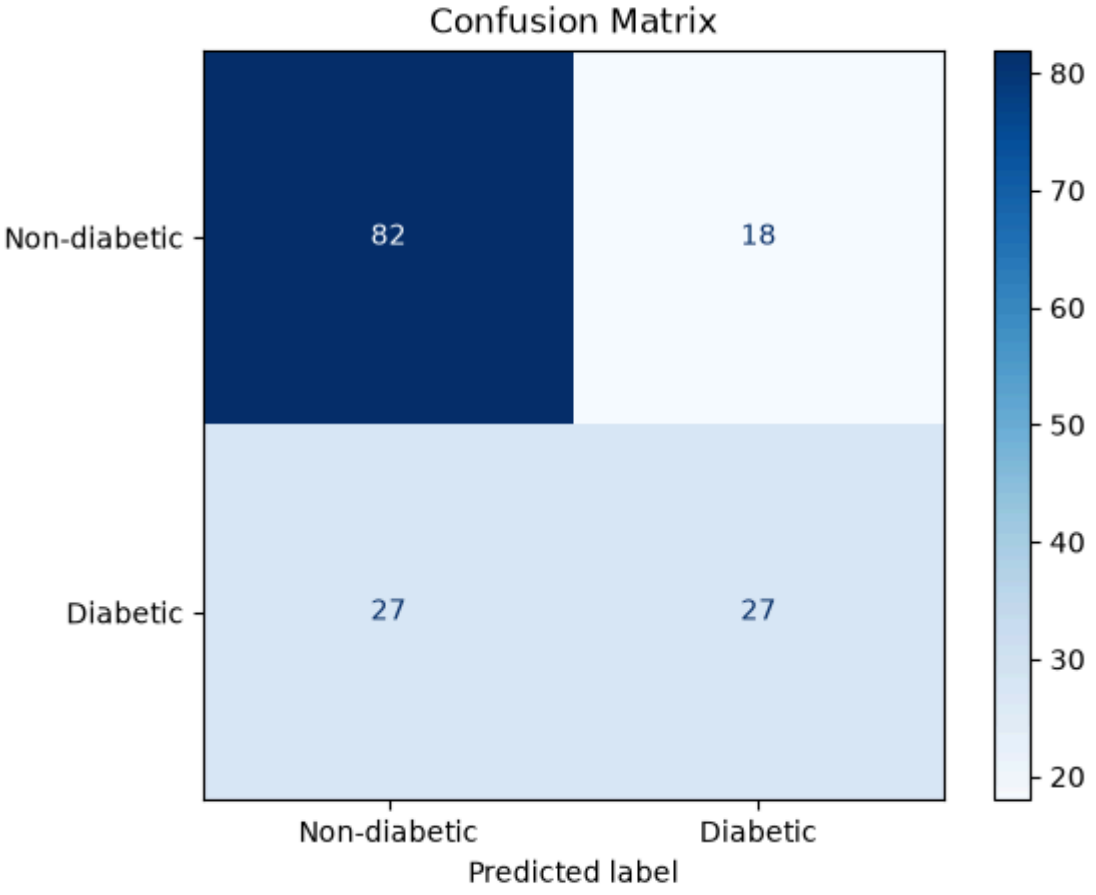
SHAP was used to identify the features that had the
greatest influence on model predictions.

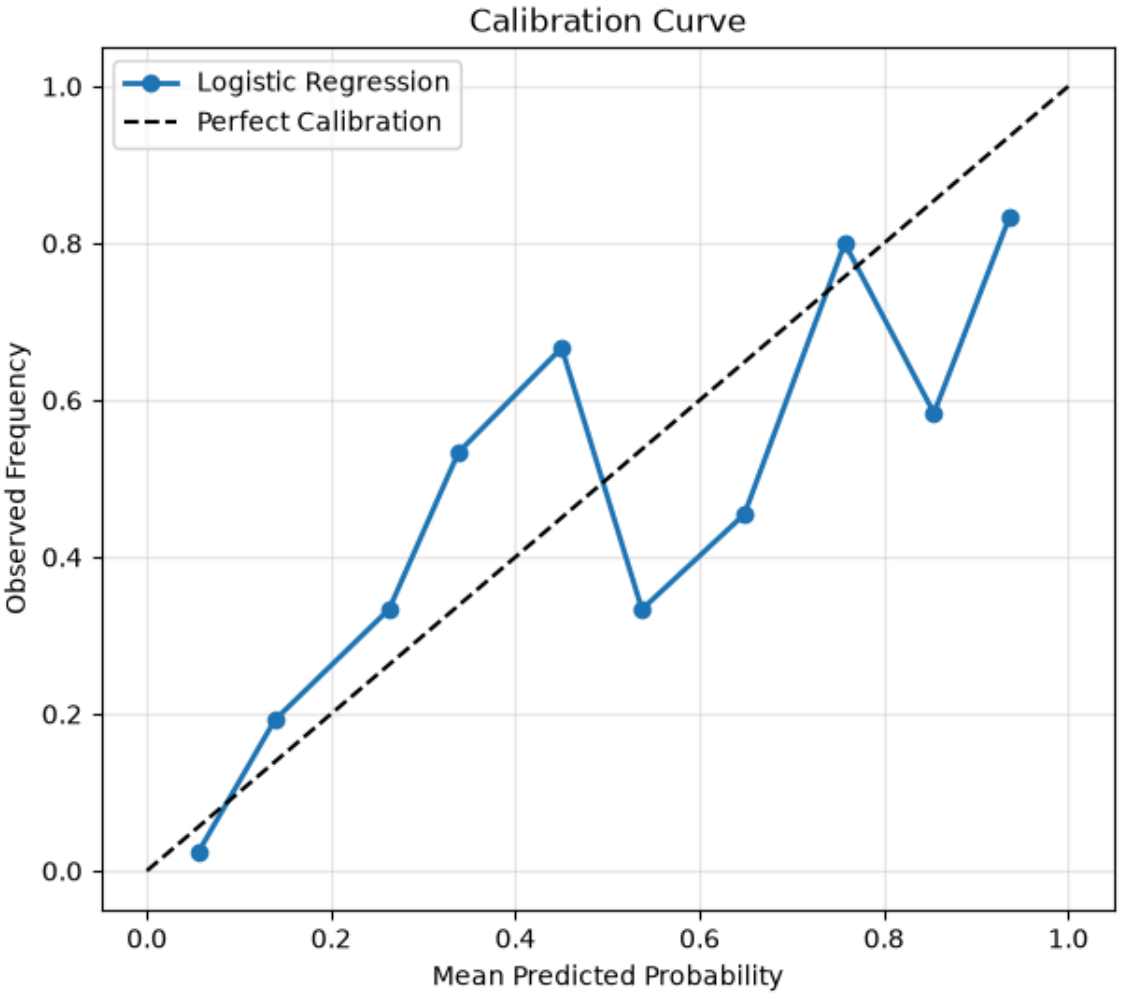
LIME was used to provide a local explanation for
an individual patient prediction.

The experiment demonstrates that machine learning
evaluation should consider not only classification
performance but also probability calibration and
model explainability.
""")

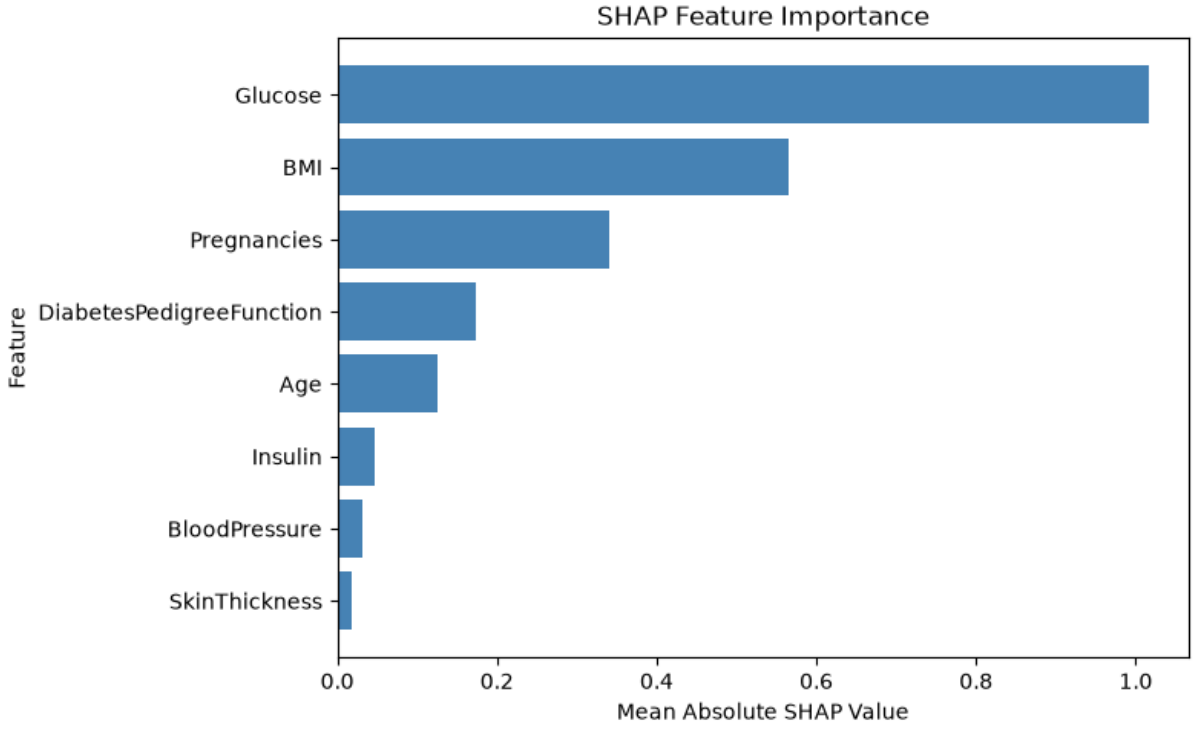
print("\nExperiment completed successfully!")
```

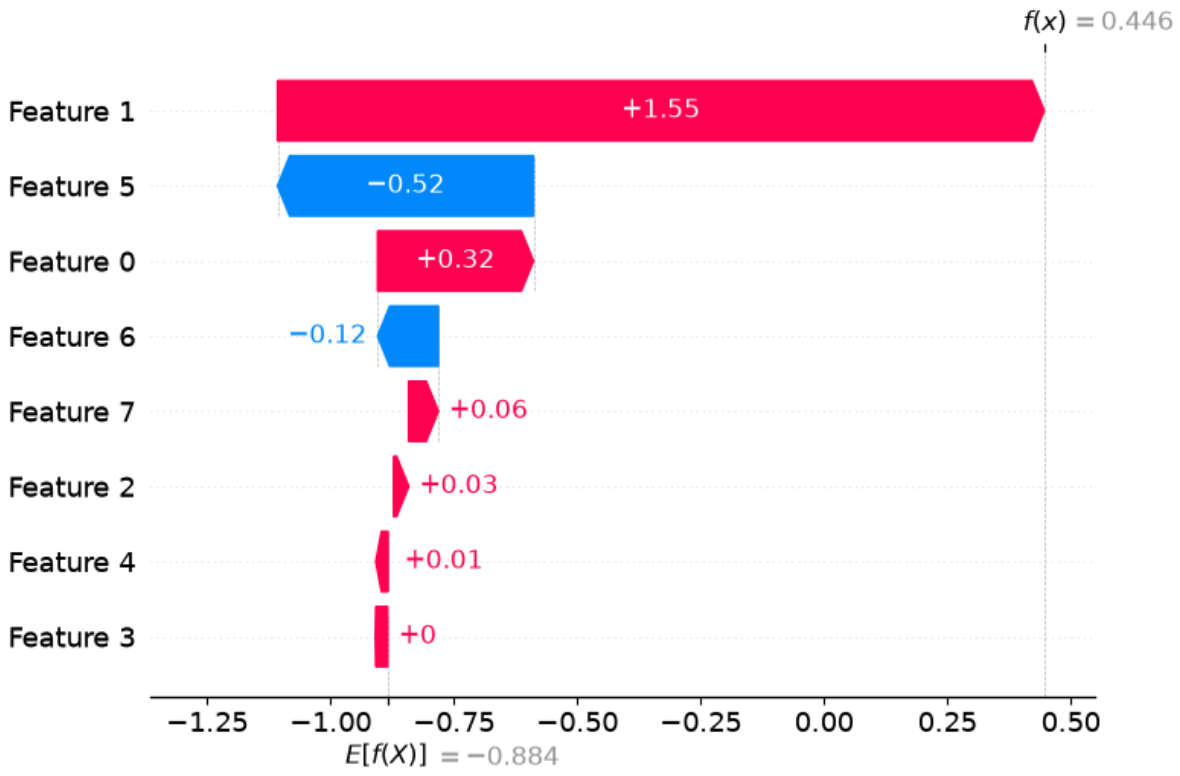
Output:-


Confusion Matrix
Non-diabetic
Diabetic
82
18
27
27
Non-diabetic
Diabetic
Predicted label
80
70
60
50
40
30
20


Calibration Curve
Logistic Regression
Perfect Calibration
Observed Frequency
Mean Predicted Probability
0.0
0.2
0.4
0.6
0.8
1.0


SHAP Summary Plot
Feature 1
Feature 5
Feature 0
Feature 6
Feature 7
Feature 4
Feature 2
Feature 3
−3
−2
−1
0
1
2
3
SHAP value (impact on model output)
High
Low
Feature value


SHAP Feature Importance
Glucose
BMI
Pregnancies
DiabetesPedigreeFunction
Age
Insulin
BloodPressure
SkinThickness
Feature
0.0
0.2
0.4
0.6
0.8
1.0
Mean Absolute SHAP Value


f(x) = 0.446
Feature 1
Feature 5
Feature 0
Feature 6
Feature 7
Feature 2
Feature 4
Feature 3
+1.55
−0.52
+0.32
−0.12
+0.06
+0.03
+0.01
+0
−1.25
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
E[f(X)] = −0.884

```
======================================================================
PIMA INDIANS DIABETES DATASET
======================================================================

Dataset Shape:
(768, 9)

First 5 Rows:
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  DiabetesPedigreeFunction  Age  Outcome
0            6      148             72             35        0  33.6                     0.627   50        1
1            1       85             66             29        0  26.6                     0.351   31        0
2            8      183             64              0        0  23.3                     0.672   32        1
3            1       89             66             23       94  28.1                     0.167   21        0
4            0      137             40             35      168  43.1                     2.288   33        1

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Pregnancies               768 non-null    int64
 1   Glucose                   768 non-null    int64
 2   BloodPressure             768 non-null    int64
 3   SkinThickness             768 non-null    int64
 4   Insulin                   768 non-null    int64
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64
 8   Outcome                   768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB

Class Distribution:
Outcome
0    500
1    268
Name: count, dtype: int64
```

```
Statistical Summary:
       Pregnancies     Glucose  BloodPressure  ...  DiabetesPedigreeFunction         Age     Outcome
count   768.000000  768.000000     768.000000  ...                768.000000  768.000000  768.000000
mean      3.845052  120.894531      69.105469  ...                  0.471876   33.240885    0.348958
std       3.369578   31.972618      19.355807  ...                  0.331329   11.760232    0.476951
min       0.000000    0.000000       0.000000  ...                  0.078000   21.000000    0.000000
25%       1.000000   99.000000      62.000000  ...                  0.243750   24.000000    0.000000
50%       3.000000  117.000000      72.000000  ...                  0.372500   29.000000    0.000000
75%       6.000000  140.250000      80.000000  ...                  0.626250   41.000000    1.000000
max      17.000000  199.000000     122.000000  ...                  2.420000   81.000000    1.000000

[8 rows x 9 columns]

Missing Values After Preprocessing:
Pregnancies                 0
Glucose                     5
BloodPressure              35
SkinThickness             227
Insulin                   374
BMI                        11
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64

Training Set Size: (614, 8)
Testing Set Size: (154, 8)
```

```
========================================================================
5-FOLD CROSS-VALIDATION
========================================================================

Accuracy:
Fold 1 : 0.7886
Fold 2 : 0.7886
Fold 3 : 0.8211
Fold 4 : 0.7805
Fold 5 : 0.7623

F1 Score:
Fold 1 : 0.6389
Fold 2 : 0.6667
Fold 3 : 0.6944
Fold 4 : 0.6494
Fold 5 : 0.6234

Mean Accuracy: 0.7882
Accuracy Std: 0.0191

Mean F1 Score: 0.6545
F1 Std: 0.0244

========================================================================
TEST SET PERFORMANCE
========================================================================

Accuracy: 0.7078
F1 Score: 0.5455

Classification Report:
              precision    recall  f1-score   support

Non-diabetic       0.75      0.82      0.78       100
    Diabetic       0.60      0.50      0.55        54

    accuracy                           0.71       154
   macro avg       0.68      0.66      0.67       154
weighted avg       0.70      0.71      0.70       154


Confusion Matrix:
[[82 18]
 [27 27]]
```

```
============================================================
BRIER SCORE
============================================================

Brier Score: 0.1754

============================================================
SHAP EXPLAINABILITY
============================================================
Background dataset has 614 samples but max_samples=100. Subsampling to 100 samples for SHAP value computation. To use all samples, set max_samples=614 when initializing the masker.

SHAP Feature Importance:
                    Feature  Mean Absolute SHAP
                    Glucose            1.015598
                        BMI            0.566222
                Pregnancies            0.341236
DiabetesPedigreeFunction            0.173953
                        Age            0.126328
                    Insulin            0.046597
              BloodPressure            0.031183
              SkinThickness            0.017992

============================================================
INDIVIDUAL PATIENT PREDICTION
============================================================

Patient Features:
    Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  DiabetesPedigreeFunction  Age
44            7    159.0           64.0            NaN      NaN  27.4                     0.294   40

Actual Outcome: 0
Predicted Outcome: 1
Predicted Probability: 0.6097

Generating SHAP individual explanation...
```

```
============================================================
LIME EXPLAINABILITY
============================================================

LIME Explanation:
Glucose > 0.61 : 0.4332
BMI <= -0.71 : -0.1773
Pregnancies > 0.66 : 0.1283
-0.70 < DiabetesPedigreeFunction <= -0.29 : -0.0306
-0.37 < Age <= 0.65 : 0.0153
-0.22 < Insulin <= -0.16 : 0.0145
BloodPressure <= -0.66 : 0.0137
-0.45 < SkinThickness <= -0.00 : 0.0042

============================================================
FINAL RESULTS
============================================================
               Metric    Value
5-Fold Mean Accuracy 0.788231
 5-Fold Accuracy Std 0.019057
      5-Fold Mean F1 0.654545
       5-Fold F1 Std 0.024423
       Test Accuracy 0.707792
             Test F1 0.545455
         Brier Score 0.175420
```

Conclusion

The classification model was evaluated using cross-validation and calibration analysis to understand its performance and reliability. Explainable AI techniques such as SHAP or LIME were applied to identify the features influencing individual predictions. The experiment demonstrated that model performance, probability reliability, and interpretability are important aspects of evaluating machine learning systems.

Questions to be Answered by Students

1.What was the mean and standard deviation of the model performance across the cross-validation folds?

Using 5-fold cross-validation:

- Mean Accuracy = 0.7882
- Accuracy Standard Deviation = 0.0191
- Mean F1-score = 0.6545
- F1-score Standard Deviation = 0.0244

The model achieved a mean accuracy of 78.82% ± 1.91% and a mean F1-score of 65.45% ± 2.44% across the five cross-validation folds. The relatively small standard deviations indicate that the model's performance was reasonably consistent across the different folds.

2.What was the obtained Brier Score, and what does it indicate about the reliability of the predicted probabilities?

Brier Score = 0.1754

The obtained Brier Score was 0.1754. The Brier Score measures the difference between the predicted probabilities and the actual outcomes, with values closer to 0 indicating more accurate probabilistic predictions. Thus, a score of 0.1754 indicates that the model's probability predictions have a moderate level of accuracy, although they are not perfect.

The Brier Score should be interpreted together with the calibration curve rather than by itself.

3.Was the model well calibrated based on the calibration curve? Justify your answer.

The calibration curve should be used to answer this directly. The numerical output you provided does not include the calibration-curve coordinates, so the graph itself is needed for a definitive statement.

For your report, the appropriate interpretation is:

The model's calibration was assessed by comparing the calibration curve with the diagonal line representing perfect calibration. A model is well calibrated when its predicted probabilities closely follow the diagonal. Therefore, if the plotted curve stays reasonably close to the diagonal, the model can be considered reasonably calibrated; substantial deviations indicate that the predicted probabilities are less reliable.

4.Which features were identified as the most influential using SHAP or LIME?

Your actual SHAP results give this ranking:

Rank	Feature	Mean Absolute SHAP
1	Glucose	1.015598
2	BMI	0.566222
3	Pregnancies	0.341236
4	DiabetesPedigreeFunction	0.173953
5	Age	0.126328
6	Insulin	0.046597
7	BloodPressure	0.031183
8	SkinThickness	0.017992

The SHAP analysis identified Glucose as the most influential feature, followed by BMI,

Pregnancies, DiabetesPedigreeFunction, and Age. Glucose had the largest mean absolute SHAP value (1.015598), indicating that it had the greatest influence on the model's predictions among the features analyzed.

SHAP importance indicates how strongly the model uses a feature; it does not by itself establish that the feature causes the outcome.

5.Explain one individual prediction using the obtained explainability results.

The selected patient had:

- Pregnancies = 7
- Glucose = 159
- Blood Pressure = 64
- BMI = 27.4
- Diabetes Pedigree Function = 0.294
- Age = 40
- Actual Outcome = 0 (Non-diabetic)
- Predicted Outcome = 1 (Diabetic)
- Predicted probability = 0.6097, or 60.97%